2. DETAILED RESEARCH PROPOSAL

2.1 Objectives (Aims)

The Canada Excellence Research Chair (CERC) program led by Dr. Nakagawa aims to transform research in ecology and evolution through a structured, inclusive, and interdisciplinary approach. Over the eight-year period, by establishing the Centre for Open Science and Synthesis in Ecology and Evolution, the program will create a lasting impact by promoting rigor, transparency, and collaboration in ecology and evolution, especially focusing on evolutionary ecology. Beyond this period, the program's influence will extend globally, offering an accessible and sustainable framework for open science and synthesis in ecology, evolution, and different fields such as social and medical sciences. The CERC program has three broad aims.

2.1.1 **Aim 1:** Create a Research Synthesis Hub for Ecology and Evolution

One of the main aims of this CERC program is to establish a research synthesis hub that serves as a knowledge and evidence nexus for ecological and evolutionary research. Under this aim, the CERC team will focus on: 1) creating a knowledge map by synthesising reviews (e.g., meta-analyses and datasets), 2) developing new synthesis methods, and 3) conducting new research synthesis including meta-(meta-)analyses, systematic mapping and bibliometric analyses. Beyond the eight-year term, this synthesis hub will continue to evolve and lead the creation of relevant syntheses and their methodologies.

2.1.2 **Aim 2:** Establish a Meta-Research Hub for Ecology and Evolution

The program's second aim is to develop a dedicated meta-research hub focused on improving the rigor and replicability of research in ecology and evolution. Through this aim, the CERC team will work on 1) quantifying and addressing biases within these fields, including the issues of equity, diversity and inclusiveness (EDI), 2) replicating/updating key meta-analyses in ecology and evolution, and 3) quantifying the replicability of key empirical studies in evolutionary ecology. This hub will serve as both a research centre and a training ground, hosting Dual Education and Research (DEAR) workshops. These workshops will bring together early-career researchers (ECRs) and established experts to conduct and teach best practices in meta-analyses, fostering a culture of transparency and replicability. Over time, the hub's impact will expand as it encourages researchers to adopt and refine meta-research methods, promoting long-term improvements in research integrity.

2.1.3 **Aim 3:** Build Infrastructure to Support Open Science and Synthesis in Ecology and Evolution

The CERC team will create a robust infrastructure that supports open science and synthesis within ecology and evolution. The key activities will include: 1) developing a big team science platform, 2) creating a semi-automated AI-based synthesis workflow, 3) developing reporting guidelines, working with relevant communities, and 4) creating the first pre-registration repository specific to ecology, evolution, and related fields. These efforts will lay the groundwork for a global shift toward collaborative, open science in ecology and evolution, with a legacy extending beyond the program's term.

2.1.4 Long-Term Vision and Impact

Beyond the initial CERC term, this program aims to establish a lasting legacy by embedding the best practices into the global research community. Dr. Nakagawa envisions a continued evolution of the synthesis and meta-research hubs, supporting international collaborations that address pressing ecological and environmental questions. The program's infrastructure will remain accessible to

researchers worldwide, promoting open science and rigorous research methodologies. Through these initiatives, the CERC program will leave an enduring impact on the scientific community, fostering a more transparent, reliable, and collaborative approach to research in ecology, evolution, and beyond.

2.2 Research Strategies and Key Activities with Methodology

2.2.1 **Aim 1:** Create a Research Synthesis Hub for Ecology and Evolution

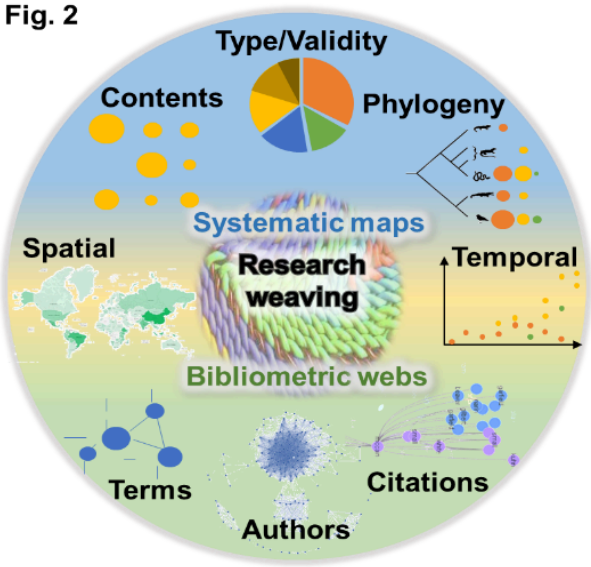
Dr Nakagawa will establish a research synthesis hub by aggregating relevant synthesis research, focusing on evolutionary ecology (within the CERC funding period) and other fields (within and beyond the CERC funding period), developing relevant methods and conducting new syntheses using cutting-edge methodology.

Aim 1.1: Mapping reviews, meta-analyses and datasets in ecology and evolution

The CERC team will systematically and bibliometrically map narrative reviews, meta-analyses, and relevant dataset papers in evolutionary ecology, creating a comprehensive database that evaluates their quality and risk of bias. Recognising the growing number of dataset publications, the team aims to collate these datasets alongside reviews and research synthesis articles to provide a holistic overview of the field. By incorporating dataset papers, they will map existing knowledge and the underlying data resources, which are crucial for advancing research synthesis and enabling secondary analyses.

To achieve this, they will employ research weaving (Fig 2). This methodology synergistically combines systematic mapping and bibliometrics to summarise information content, history, and networks among a collection of documents on any given topic. Systematic mapping will offer insights into the current state of knowledge and evidence, identifying areas with research gaps and clusters, including where datasets are available or lacking. Complementing this, bibliometrics will reveal how pieces of evidence and datasets are interconnected, highlighting the structure and development of the field, key researchers, and the influence of their work[1]. By integrating visual summaries and focusing on dynamic, interactive representations (as illustrated in Fig. 2), Dr. Nakagawa and the CERC team will make research results and data more accessible to stakeholders and the public. The comprehensive mapping of reviews, often referred to as an "overview of reviews" or a "second-order synthesis," will not only catalogue existing knowledge and datasets but also evaluate their quality and biases. This approach lays a robust foundation for identifying knowledge gaps, informing future research directions, and fostering collaboration within the scientific community by highlighting both the intellectual and data resources available for advancing evolutionary ecology.

Fig. 2



Aim 1.2: Develop new quantitative and qualitative methods for research synthesis

Recognizing that current methods may not fully capture the complexity and nuances of ecological data, the CERC team will innovate and extend statistical techniques to enhance the rigor and scope of meta-analyses and systematic reviews[2]. This includes creating new statistical and synthesis tools that allow for more comprehensive comparisons and syntheses of research findings across diverse studies.

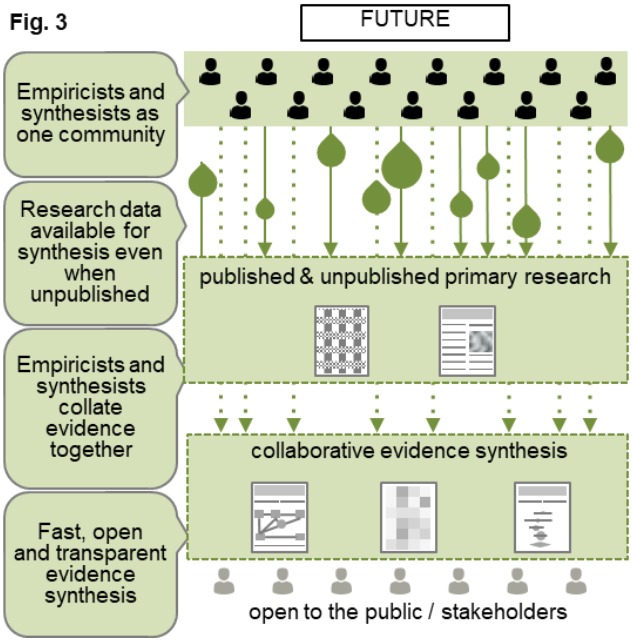
One key example of this initiative is the development, implementation, and validation of a new set of effect sizes for comparing differences in covariance, skewness, and kurtosis between two groups (e.g., females vs. males, treatment vs. control). These advanced effect sizes will enable meta-analyses to delve deeper into the distributional characteristics of data, moving beyond mean differences to understand variability, asymmetry, and the relationships between traits. Such tools are particularly valuable for studying sex differences in non-human organisms, shedding light on the co-selection of traits, disease comorbidities, and sex-biased diseases (e.g. [3-6]). While the focus is on non-human studies, these methods are also applicable to human research on both sex and gender differences.

Beyond this example, the CERC team will explore other innovative methods to enhance research synthesis, such as the methodology for meta-meta-analyses where the non-independence issue is unresolved. Building on Dr. Nakagawa's former contributions—such as introducing statistical methods tailored to ecology and evolution[7], developing phylogenetic multilevel meta-analyses[8], and creating tools to address publication bias[9]—the team will continue to push methodological boundaries. They will work on improving graphical representations of meta-analytic results (e.g., orchard plots) and integrating advanced computational techniques like machine learning and AI into synthesis workflows (see Aim 3.2).

Aim 1.3: Conduct new research syntheses where gaps in our knowledge exist

Dr. Nakagawa and the CERC team will conduct new research syntheses in areas where significant gaps have been identified through the comprehensive mapping in Aim 1.1. Given that many topics in evolutionary ecology need more synthesis, the team aims to fill these voids by initiating collaborative and innovative synthesis projects. A pivotal component of this effort is the creation of the first open synthesis community[10] (Fig 3), designed to enable a continuously updated meta-analysis in ecology and evolution.

Fig. 3



This open synthesis community will unite both empiricists—researchers conducting primary studies—and synthesists—those specializing in research synthesis. By fostering a collaborative environment, researchers will perceive their individual work as integral pieces of a larger puzzle[11,12]. This collective approach has several advantages: it ensures all results contribute to the synthesis regardless of novelty or significance, thereby mitigating publication bias; it promotes inclusivity and diversity by welcoming contributions from researchers globally, including those from underfunded regions or institutions; and it keeps syntheses up to date by allowing empiricists to directly upload their findings and receive recognition through contributorship[13]). The CERC team will consult the ecological and evolutionary biology community on what topic should be tackled first by the open synthesis community[10].

To broaden the impact and foster community engagement, the CERC team will integrate citizen science activities, particularly targeting high school students through educational outreach and school visits. By involving students in the scientific process and highlighting the importance of evolutionary studies, the

project aims to enhance STEM participation and inspire the next generation of scientists. Additionally, the team will collaborate with local teachers and researchers to co-create educational materials based on examples of local evolutionary ecology, ensuring that the outreach is both locally relevant and culturally sensitive.

2.2.2 **Aim 2:** Establish a Meta-science Hub for Ecology and Evolution

The CERC team will extensively examine various ecological and evolutionary research biases, including equity, diversity, and inclusion in ecology and evolution. By identifying these biases, the team seeks to improve the quality and inclusiveness of scientific inquiry. Through the innovative workshops, the Centre will facilitate the updating of existing meta-analyses. Additionally, the Centre will focus on quantifying the replicability of key empirical studies in evolutionary ecology.

Aim 2.1: Elucidate different kinds of biases in ecology and evolution, including EDI issues

The CERC team will systematically investigate various biases that affect research in ecology and evolution, with a particular focus on equity, diversity, and inclusion (EDI) issues. Using the comprehensive database of reviews and meta-analyses compiled in Aim 1.1, the team will employ advanced statistical techniques to quantify statistical biases such as replicability, lack of power, Type M (magnitude) errors, and Type S (sign) errors across the field. By assessing the prevalence of these errors, as outlined in recent studies)[14,15], the team aims to understand how these biases influence the reliability and validity of research findings in ecology and evolution.

In addition to statistical biases, the team will explore EDI-related biases by examining patterns in authorship demographics, geographic representation, and the accessibility of research outputs (e.g., [16,17]). The CERC team will analyze the extent to which underrepresented groups are included in research activities and publications, identifying potential disparities and barriers. This analysis will involve bibliometric studies to map collaboration networks and authorship trends, revealing systemic issues such as gender imbalances, underrepresentation of researchers from developing countries, and biases in topic selection.

To gain deeper insights into the barriers researchers face in open science, and the prevalence of questionable research practices (QRPs), the CERC team will conduct a comprehensive survey of the ecology and evolution research community. Drawing inspiration from the methodology of Fraser and colleagues[18] (Dr Nakagawa was part of the team), the survey will collect data on researchers' experiences with QRPs, perceptions of replication studies, and challenges related to research transparency and openness. The survey will be distributed widely to capture diverse perspectives across career stages, institutions, and geographic regions.

The team will identify specific factors contributing to biases and QRPs in the field by combining quantitative data from meta-analyses with qualitative insights from the community survey. This dual approach allows for a holistic understanding of both the statistical underpinnings of bias and the socio-cultural factors influencing research practices. The findings will shed light on how biases may disproportionately affect certain groups, thereby informing targeted interventions to promote EDI.

Based on the results, Dr. Nakagawa and the CERC team will develop and propose actionable strategies to mitigate identified biases and enhance research integrity. These strategies may include advocating for increased transparency through open data and open methods, promoting the adoption of best practices in study design and reporting (see Aim 3.3), and implementing training programs focused on responsible conduct of research. Collaborations with professional societies, journals, and funding agencies will be pursued to encourage policy changes that support diversity, inclusivity, and rigorous research practices.

Aim 2.2: Update (replicate) meta-analyses using innovative workshops

The main focus of this aim is to update and replicate influential meta-analyses in ecology and evolution, identified through Aim 1.1 (target of 20 meta-analyses). Given that many existing meta-analyses may have utilised outdated or suboptimal methods, the CERC team will reanalyze these studies using cutting-edge techniques such as multilevel phylogenetic meta-analyses. This will enhance the reliability and relevance of key findings by incorporating new data and advanced methodologies.

To achieve this goal, the team will employ an innovative, collaborative approach through Dual Education and Research (DEAR) Workshops (planned to be hosted annually after the first year). Each workshop will update three meta-analyses involving experts and early-career researchers (ECRs), who will work together to update a specific meta-analysis.

Equity, Diversity, and Inclusion (EDI) will be central to these workshops. They will be advertised broadly, including through networks aimed at increasing representation from systematically underrepresented groups. The workshops will strive for global representation by encouraging participation from researchers in the Global South. To facilitate this, subsidies will be offered to scholars from developing nations (based on World Bank Income Categories), and virtual participation options will be provided [19,20]. All workshops will offer accessible options for individuals with caregiving responsibilities, disabilities, or other barriers to in-person attendance. Also, Dr. Nakagawa plans to host ECRs, especially from the Global South, who wants to learn synthesis methods as a part of the Centre's visitorship program (~2-3 month visit per visitor).

The CERC team aims to assess how results have evolved due to methodological advancements and new data by reanalysing key meta-analyses with advanced methods and inclusive collaboration. Comparing updated results with original findings will provide valuable insights into the robustness of previous conclusions. This process not only advances the field by refining existing knowledge but also fosters capacity building and professional development among ECRs.

Aim 2.3: Quantify the replicability of key empirical studies in evolutionary ecology

The CERC team aim to assess the replicability of foundational empirical studies in evolutionary ecology to strengthen the reliability of the field's scientific knowledge. To select these key studies, they will consult popular textbooks on evolutionary biology and behavioral ecology, as well as identify highly cited papers through the bibliometric analyses conducted in Aim 1.1 (e.g., [21-25]). This approach ensures that the studies chosen are both influential and representative of critical concepts within the discipline.

Given technologies to access replicability is rapidly evolving, the team will employ various methods to estimate replicability at a high level. These may include aggregating expert opinions, using artificial intelligence algorithms designed to predict study reliability, and analyzing methodological indicators such as sample sizes and statistical power (e.g., [26,27]). By combining different approaches, the team seeks to provide a comprehensive assessment without relying on a single method. This effort will highlight areas where empirical findings are robust or where further verification is needed, ultimately contributing to advancing evolutionary ecology by promoting transparency and encouraging rigorous research practices.

2.2.3 Aim 3: Build Infrastructure to Support Open Science and Synthesis in Ecology and Evolution

The CERC team aims to develop essential infrastructure to facilitate open science and collaborative research in ecology and evolution. This includes creating an online collaborative platform to enable big-team science, developing a semi-automated AI-based synthesis workflow, producing guidelines to encourage best practices, and building the first modular registration system

Aim 3.1: Develop a computational platform that enables big-team science

The CERC team will develop a cutting-edge online collaborative platform designed to facilitate big-team science by implementing Co-authorship-Incentivized Mega-Projects (CIMPs). CIMPs represent an innovative, collaborative model that leverages registered reports. In this emerging publishing format, project ideas and methodologies are peer-reviewed and accepted for publication before data collection and analysis. This approach ensures that participation is encouraged based on the scientific merit of the project rather than the novelty or statistical significance of the results. The team's successful implementation of a CIMP in a many-analyst project, which engaged over 220 scientists contributing an estimated 4,000 person-hours, demonstrates the efficacy of this model in fostering large-scale collaboration and enhancing research reproducibility[28] (cf. [29]).

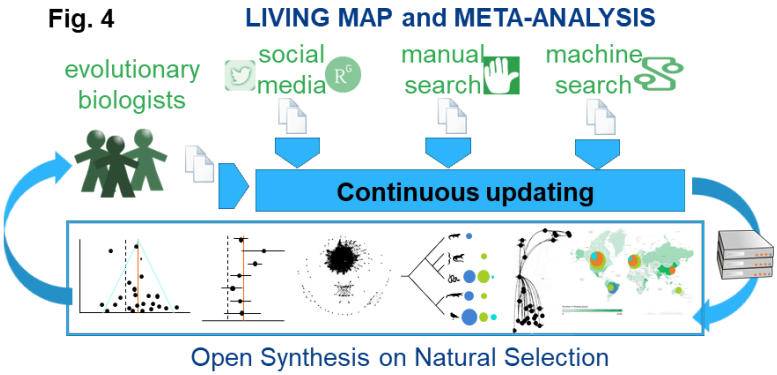
The planned online collaborative platform will be the backbone for CIMPs, providing the necessary infrastructure to manage and coordinate extensive collaborative efforts. The platform's key features will include integrated project management tools, real-time data-sharing capabilities, and streamlined communication channels to support interaction among participants. Additionally, the platform will incorporate functionalities for submitting and tracking registered reports (or pre-registration), ensuring that all stages of the research process are transparent and accessible. By automating administrative tasks and facilitating efficient workflow management, the platform will enable researchers to focus on scientific inquiry and data analysis, accelerating the pace of discovery in ecology and evolution.

Furthermore, the platform will enhance inclusivity and diversity in research by supporting the CIMP model's co-authorship incentives. By providing a centralized space where volunteer scientists can contribute their expertise to various datasets and participate in the evaluation of peer analyses, the platform will attract diverse participants from different geographical regions and institutional backgrounds. Features such as virtual collaboration tools and accessibility options will ensure that researchers facing barriers to in-person participation can engage fully in CIMPs.

Aim 3.2: Create a semi-automated AI-based synthesis workflow with living synthesis capacity

This workflow will integrate artificial intelligence and machine learning technologies to semi-automate key components of systematic reviews and meta-analyses, such as literature screening, data extraction, and preliminary data analysis. By leveraging AI, the team aims to significantly reduce the time and resources required for comprehensive research syntheses, enabling more frequent and extensive analyses. This approach not only streamlines the synthesis process but also minimizes error and increases the consistency of data handling, thereby improving the overall quality of meta-analytic outcomes.

Fig. 4



Central to this aim is the establishment of living synthesis capacity, which refers to maintaining continuously updated meta-analyses that incorporate new research findings in real-time[30,31](Fig 4, an example of a workflow for a hypothetical project). The AI-based workflow will facilitate this by automatically monitoring relevant databases and journals for new studies, extracting pertinent data, and updating existing meta-analytic models accordingly. This dynamic updating process ensures that syntheses remain current and reflect the latest scientific developments, providing researchers,

policymakers, and stakeholders with up-to-date evidence for decision-making. Additionally, the workflow will include features for version control and transparency, allowing users to track changes and understand the evolution of the synthesis over time. By implementing living syntheses, the CERC team will foster an environment of ongoing knowledge integration and adaptability, essential for addressing rapidly advancing and complex ecological and evolutionary questions. Furthermore, later in the program, Dr. Nakagawa aims to integrate the semi-automated AI-based synthesis workflow with the computational platform in Aim 3.1, enabling big-team science through enhanced collaboration and data sharing.

Aim 3.3: Produce guidelines for best practices in research syntheses in ecology and evolution

The CERC team will develop Community-Driven and Dynamic (CDD) guidelines to establish and promote best practices in research synthesis within ecology, evolution, and related fields. Building on the success of the Preferred Reporting Items for Systematic reviews and Meta-Analyses for Ecology and Evolution (PRISMA-EcoEvo)[32], which was collaboratively created with the original authors of PRISMA, the team will continue to engage stakeholder communities in the creation and continuous updating of these guidelines and new guidelines which the CERC team will initiate. This collaborative approach ensures that the guidelines are scientifically robust, widely accepted, and owned by the research community, thereby fostering widespread adoption and adherence.

To achieve this, the CERC team will form a dedicated community of researchers, practitioners, and other stakeholders who will actively participate in developing and refining the guidelines. Aim 3.3 will focus on producing review articles and detailed guidelines that address various aspects of research synthesis, including systematic reviews, meta-analyses, and empirical studies. A key example will be the creation of a guideline specifically for updating meta-analyses in ecology and evolution, which will provide a standardized framework for researchers to follow, ensuring consistency, transparency, and methodological rigor in ongoing synthesis efforts. These guidelines will incorporate feedback from the community through workshops, surveys, and collaborative platforms, ensuring they remain relevant and responsive to the field's evolving needs. In the future, the CERC team will expand the scope of these guidelines to encompass empirical studies, promoting best practices that enhance the reliability and reproducibility of primary research.

Aim 3.4: Building the very first modular registration system to spread the culture of registration across the ecology and evolution community

The CERC team will develop the first modular registration system tailored specifically for the ecology and evolution research community[33]. This innovative system will facilitate the adoption of pre-registration practices by providing a flexible and scalable platform that accommodates various study designs and methodologies inherent to these fields. Unlike traditional pre-registration platforms, the modular system will allow researchers to select and customize registration modules that best fit their specific research needs, whether they are conducting observational studies, experiments, meta-analyses, or theoretical modeling. By offering tailored templates and guidelines, the system will lower the barriers to pre-registration, encouraging more widespread and consistent use across diverse research projects.

The implementation of this modular registration system will significantly enhance the transparency, reproducibility, and credibility of research in ecology and evolution. Researchers will be able to clearly outline their hypotheses, methodologies, and analysis plans before data collection begins, thereby reducing the risk of selective reporting and increasing the reliability of study outcomes. To promote widespread adoption, the CERC team will engage with key stakeholders, including academic journals, funding agencies, and professional societies, to advocate for the integration of the modular registration

system into standard research practices. Educational workshops and training sessions will be conducted to familiarize researchers with the system, emphasizing its benefits for enhancing research integrity and facilitating collaborative, open science working with learnt societies such as the Society for Open, Transparent Ecology and Evolutionary biology (SORTEE) and the Association for Inter-disciplinary Meta-science and Open Science (AIMOS).

EDI considerations: The CERC program will integrate EDI considerations into every facet of research, from formulating questions and designing studies to data collection, analysis, interpretation, and dissemination. By systematically mapping available data, the program enhances global representation in synthesis studies and targets data collection in underrepresented areas. Central to this effort is creating an open synthesis community that employs inclusive advertising and equitable recruitment practices to engage historically excluded and marginalized groups, thereby reducing research inequalities, improving data synthesis quality, and ensuring diverse perspectives are included. . Moreover, the program significantly expands Sex- and Gender-Based Analysis Plus (SGBA+) by developing new statistical methods to investigate sex and gender differences, addressing gaps in existing guidelines.

2.3 Leadership

UofA is internationally recognized in both relevant fields, Ecology & Evolution and Artificial Intelligence (AI); however, the two fields are not yet strongly connected. Dr. Nakagawa will become a conduit between these two areas of strength at UofA. He will nucleate an internationally unique and much needed collaborative comprising ecologists, evolutionary biologists, and computer scientists. Dr. Nakagawa's research is exceptionally cross-disciplinary, and he has been bridging different disciplines for several years by mobilizing and exchanging ideas and insights from different fields (especially ecology, evolution, statistics, data & computer sciences, psychology, and biomedicine) and collaborating with >750 scientists (see Section A). Importantly, the CERC program will intersect with four of the five signature areas of the host institution: 1) AI4Society, 2) Precision Health, 3) Research at the Intersections of Gender, and 4) Situated Knowledges: Indigenous Peoples and Place. These four signature areas are well aligned with all five of the *CERC's Science, Technology, and Innovation Priorities*. Dr. Nakagawa will also become a champion of open and reproducible science at UofA, connecting all existing advocates of open science movements at the host institution, who are scattered across different faculties. Further, the CERC program will bolster the UofA's campaign for trust in science, spearheaded by Dr. Tim Caulfield.

Although several open science and meta-science initiatives exist in Canada, such as the Centre of Journalology (Ottawa Hospital), Tanenbaum Open Science Institute (Montreal Neurological Institute), and the Living Data Project (Canadian Institute of Ecology and Evolution), Canada currently lags other countries in these fields[34]. For example, the UK Reproducibility Network (UKRN), funded in 2015 and including >25 universities, is a peer-led consortium for improving and supporting rigour and reducibility in research by providing training, peer support, and infrastructure at the local and national levels. Similarly, a new US initiative, the Higher Education Leadership Initiative for Open Scholarship, includes 75 US universities. The Center of Open Science, established in 2013, has been a leader in open and reproducible science in terms of research and training. There has been a movement led by Dr. David Moher (Ottawa) to establish the Canadian Reproducibility Network so that different Canadian institutions can cooperate to improve research quality and reproducibility concertedly. Dr. Nakagawa plans to work with Dr. Moher to establish and build such a network in Canada (note that Drs Moher and Nakagawa have already collaborated to create PRISMA-EcoEvo[32]. Of relevance, Dr. Nakagawa is a founding member of the Australian Reproducibility Network, has co-founded an international society, SORTEE, and was a former executive member of another international society, AIMOS. In terms of

research synthesis, the CERC will work closely with the Canadian Centre for Evidence-based Conservation (CCEC; directed by Dr. Steven Cooke,), with CCEC focusing on research synthesis on applied and practical issues and the CERC program on synthesizing more fundamental issues.

The CERC team will also undertake syntheses of cross-disciplinary nature. Many pressing issues facing Canada today (e.g., climate change, public trust in science, sustainability, gender/sex differences; see Aim 1.1) transcend disciplinary boundaries, yet a large gap exists for interdisciplinary syntheses that can guide research and policy. Dr. Nakagawa will play a key role in the national-level effort to fill this gap.

2.4 Training Strategies

Mentoring and developing Highly Qualified Personnel (HQP) is central to Dr. Nakagawa's academic mission. As a coach and facilitator, he fosters an environment where trainees can realize their potential as independent thinkers and innovators. His approach combines technical expertise and soft skills development in a supportive, collaborative, and inclusive environment. Dr. Nakagawa's extensive international network, built through collaborations with over 1,000 researchers across all continents, provides valuable opportunities for HQP. By connecting trainees with global collaborators, he enables them to engage in cross-disciplinary projects, broaden their perspectives, and establish professional networks. This collaborative ethos extends within his lab, where open communication, mutual respect, and continuous learning are paramount. Trainees work both vertically and horizontally, with more experienced members mentoring juniors while peers collaborate on shared projects.

Tailored training plans are developed for each trainee, addressing individual strengths, skill gaps, and aspirations. Dr. Nakagawa emphasizes skills in two key areas: advanced statistical and computational methods, and scientific communication. He regularly organizes workshops, book clubs, and writing clinics to enhance trainees' proficiency in these areas, especially aiding non-native English speakers to overcome language barriers. This commitment to inclusive training has brought trainees from over 18 countries to his lab, fostering a culturally diverse team. To further support equity, diversity, and inclusion (EDI), Dr. Nakagawa employs inclusive recruitment practices, offering flexible work arrangements to accommodate caregiving responsibilities and other needs. Weekly meetings and one-on-one sessions provide a platform for well-being checks and open dialogue. Continuous EDI training and discussions help create a cohesive team where diverse perspectives enrich research approaches.

At any point after one year, Dr. Nakagawa will have > 14 HQP, including postdoctoral fellows, PhD and MSc students, and undergraduates. His postdocs lead complex projects, mentor junior researchers, and benefit from international collaborations, while graduate students develop methodological, programming, and communication skills through hands-on research. Undergraduates serve as research assistants, gaining foundational skills and early exposure to research. Dr. Nakagawa's inclusive and holistic training model equips HQP with advanced technical skills, leadership, and global competency, preparing them for diverse careers in academia, industry, and government. The program integrates EDI at all stages, aiming to cultivate a workforce that reflects diverse backgrounds and perspectives essential to the future of ecology, evolution, and STEM fields more broadly.

2.5 Outcomes and Benefits for Canada

The CERC program led by Dr. Nakagawa will position Canada at the forefront of ecological and evolutionary research by enhancing research transparency, reliability, and collaboration. By creating dedicated synthesis and meta-research hubs, the program will elevate Canada's standing as a global leader in open and team science, fostering a more rigorous and trustworthy foundation in research across fields. This initiative will enable Canada to address critical challenges in climate change, biodiversity

conservation, and public health by equipping policymakers and stakeholders with reliable, evidence-based insights drawn from a comprehensive body of ecological and evolutionary research.

One of the primary benefits of the program is its potential to make Canadian research teams more competitive on the global stage. The synthesis hub, which will provide a continuously updated knowledge base, and the meta-research hub, which will address research biases, will strengthen Canada's capacity for high-quality research. By systematically mapping and evaluating current knowledge in ecology and evolution, the program will uncover research gaps and biases that Canadian teams can address, driving innovation and enabling Canadian scientists to take leadership roles in addressing global research challenges. This CERC program will thus establish Canada as a key player in setting standards for research transparency and reproducibility in ecology and evolution, enhancing Canada's scientific influence.

The program will also bring substantial social, cultural, and economic benefits to Canada by fostering an inclusive and collaborative research environment. Through DEAR (Dual Education And Research) workshops, the program will train early-career researchers in best practices for meta-analysis and research synthesis, expanding the country's talent pool in data-intensive ecological and evolutionary research. This training builds skills and cultivates a new generation of scientists committed to open science practices, promoting a culture of transparency and integrity in research. By embedding equity, diversity, and inclusion (EDI) considerations in every aspect of the program, this CERC initiative will help diversify Canada's scientific community, making it more representative and resilient.

Economically, the CERC program will produce a highly skilled workforce in data science and evidence synthesis, areas with growing demand in Canada and worldwide. Trainees will gain expertise in advanced statistical methods, research synthesis, and open science practices—skills that are in high demand across multiple sectors, including environmental management, conservation, and public health. This workforce development addresses a critical skills gap in Canada and will enhance the competitiveness of Canadian institutions and industries on the international stage. Furthermore, by improving research efficiency and transparency, the program will ensure that public investments in research yield greater returns, creating more value from existing datasets and informing cost-effective, evidence-based policy.

Environmentally, the CERC program will empower Canadian scientists and policymakers to make better-informed decisions on biodiversity conservation and climate adaptation. Through systematic reviews and meta-analyses, the program will provide insights into the impacts of environmental changes on ecosystems, supporting Canada's commitments to sustainable resource management and environmental resilience. By making these findings publicly accessible, the program will also increase public awareness of ecological and evolutionary processes, fostering greater appreciation and support for conservation efforts across Canada.

2.6 Knowledge Translation and Mobilization Strategy

The CERC program is committed to effectively disseminating research findings and innovative methodologies within the ecology, evolution, and related fields through a comprehensive Knowledge Translation and Mobilization Strategy. Beyond traditional academic channels—such as publishing in top journals, presenting at conferences, and developing open-source software like R packages—the team will establish two primary communication avenues to engage academics, the public, and key stakeholders, including government and not-for-profit organizations.

First, a dedicated website will be developed to serve as a central hub for all CERC activities and communities. This platform will facilitate knowledge exchange and collaboration among researchers

while making science accessible to the public. By integrating with initiatives like *EcoEvoRxiv* and *EcoEvo Registry*, the website will promote open, transparent, and reproducible science, providing a permanent channel for continuous communication and knowledge sharing within the field.

Second, the program will establish an outreach team tasked with coordinating school and community visits, as well as media interviews and broadcasts of research findings. This outreach will include the creation of educational materials, such as videos and extensive online tutorials (e.g., potentially creating a Massive Open Online Course, MOOC), aimed at enhancing public appreciation of STEM and inspiring future careers in these areas. Collaborating with the host institution's media relations, the outreach team will effectively communicate with the Canadian public and government, leveraging existing connections to maximize the program's impact. Through these initiatives, the CERC program will foster a more informed and engaged community, promote scientific literacy, trust in science, and inspire the next generation of ecologists and evolutionary biologists.

2.7 Sustainability Plan

The University of Alberta (UofA) is dedicated to ensuring the long-term sustainability of the CERC program beyond the initial eight-year funding period through a comprehensive, multi-tiered strategy. First, UofA maintains and will maintain essential research infrastructure by providing Dr. Nakagawa with continued access to office and lab spaces, dedicated conference/meeting rooms, and core facilities. Future upgrades and acquisitions of new research equipment will be supported through institutional budgets and major federal infrastructure programs such as the Canada Foundation for Innovation (CFI) and Natural Sciences and Engineering Research Council (NSERC) Research Tools and Instruments (RTI) grants. The existing strengths in Ecology & Evolution and AI at the UofA, bolstered by a substantial number of tenured and tenure-track faculty, alongside Dr. Nakagawa's high international profile, will enhance the team's ability to secure these competitive funding sources.

Second, the UofA will provide ongoing support for CERC-aligned researchers through faculty positions and robust research funding. The Faculty of Science's strategic focus on areas including ecology, wildlife biology, conservation, biodiversity, mathematical biology, AI, data mining, and open science aligns well with the CERC's objectives, ensuring continued investment in these priority areas. The planned hire of an additional faculty member in "meta-science and open science in ecology and evolution" further demonstrates the UofA's commitment to building capacity and ensuring the program's longevity. Graduate students and postdoctoral fellows associated with the CERC will receive sustained support through departmental teaching assistantships, NSERC Discovery Grants, and the establishment of a subsequent chair position, embedding the program deeply within the university's research ecosystem.

Lastly, the sustainability of the CERC program will be reinforced through strategic partnerships and collaborative networks. By leveraging relationships with research networks such as Amii, ABMI, ACA, Alberta Environment and Parks, Cybera, Environment and Climate Change Canada, and the Global Ecological Integrity Group, the CERC team will secure ongoing support for data acquisition, curation, and research infrastructure. Additionally, UofA's commitment to establishing an endowed chair position for Dr Nakagawa (also with a nomination to a tier-1 CRC) upon the conclusion of the CERC, alongside fostering lasting partnerships with commercial entities and policymakers, will ensure the continued impact and growth of the program. These efforts, combined with the development of open synthesis communities and AI-based research tools, will create a robust foundation for the CERC program to thrive well beyond the initial funding period, driving forward scientific discovery and maintaining leadership in open, reproducible, and inclusive research practices in ecology and evolution.

2.8 References

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